The target box: the strong-coupling gap survives local perturbations with an existential tolerance; its explicit location is open

Correction (2026-10-02). This note inherits the 2026-10-02 correction of the Kogut–Susskind note, which withdrew that note's volume-uniform thresholds. The changes here:

1. The former Proposition, "for $\lambda = (1-\theta)\varepsilon'$, $u' \le \frac{1}{2}$ and $4Fu' \le \lambda$, every $H \in \mathcal{C}(g,\eta,\eta_E,q)$ has a unique ground state and a gap of at least $\frac{8}{3}g^2(1-\eta_E)(1-\theta)\,\hbar c/a$, uniformly in the volume", is **withdrawn**. Its criterion was the Kogut–Susskind one, whose adjacent count misses histories that reach the test link after their first insertion and whose general count is withdrawn. That criterion addressed decay on a cyclic subspace of local observables at best, and no true-vacuum correlation bound was supplied. §1 replaces the Proposition by an existential one, proved by a hypothesis check of Yarotsky's theorem for translation-invariant perturbations; §§2–3 keep the explicit count as a formal estimate.
2. The former lead table was offered as a tolerance table. Its $q = 4$ column contradicted the note's own criterion $e\,\eta\,2^q \le 0.15g^2 - 384e/g^2$: the printed 0.29 and 0.62 at $g^2 = 100, 200$ should read 0.104 and 0.569 (the printed 1.3 at 400 agrees). The $q = 8$ entries 0.0066 and 0.036 were rounded up from 0.00655 and 0.0356. For $q > 4$ the corrected overlap multiplicity adds a condition that binds for $q = 8$ at $g^2 \ge 200$ (§3). The first-step factor $F = 8Ne/g^2 + e\eta$ omitted the 2^q support outcomes of a first d-insertion. §3 gives the corrected values, as formal estimates.
3. The remark "the rigorous general count multiplies the per-step factor by 24, so the same table holds at 24 times the coupling, $g^2 = 2400, 4800, 9600$, or at the listed couplings with tolerances divided by 24" is withdrawn with that count. Its second reading was also false as algebra (§3).
4. The finish line "g^2 of order 10^2 with few-link corrections of local norm below a few per cent of $\hbar c/a$" is a formal estimate. The proved box of §1 has an existential location.

The strong-coupling gap of the $SU(3)$ Kogut–Susskind Hamiltonian survives bounded, gauge-invariant, translation-invariant few-link perturbations and a link-local deformation of the electric term, uniformly in the volume. Precisely (§1), let $\mathcal{C}_{\mathrm{TI}}(g,\eta,\eta_E,q)$ be the class of Hamiltonians

$$H = H'_E - \frac{1}{g^2}\sum_p w_p - \sum_j d_j$$

on a periodic cubic lattice of side $N_s \ge 2q+1$, in units $\hbar c/a$. Here H'_E is link-local and diagonal in the Peter–Weyl basis, and every nontrivial link representation costs at least $(1-\eta_E)\varepsilon$ with $\varepsilon = 2g^2/3$ and $0 \le \eta_E < 1$. Each d_j is bounded, self-adjoint, gauge invariant and supported on a connected set of at most q links, the family $\{d_j\}$ is translation covariant, and $\sum_{j\ni\ell}\|d_j\| \le \eta$ for every link ℓ. If

$$\eta \;\le\; \frac{2(1-\eta_E)\beta_q^*}{9}\,g^2 - \frac{6}{g^2},$$

then H has a non-degenerate, gauge-invariant ground state, and its full and physical gaps are at least

$$\gamma_q\,(1-\eta_E)\,\frac{2g^2}{3}\,\frac{\hbar c}{a}$$

for every such N_s. The constants $\beta_q^*, \gamma_q > 0$ are those of Yarotsky's Theorem 1 for the range $\{-q,\dots,q\}^3$; they exist and are not known as numbers. The tolerance grows linearly in g^2 with an existential slope, and the box opens at $g^4 \ge 27/[(1-\eta_E)\beta_q^*]$.

That turns the strong-coupling result into a **target**: a renormalization scheme that is an exact low-energy reduction and steers the effective Hamiltonian into $\mathcal{C}_{\mathrm{TI}}$ inside this region gives a volume-uniform lattice gap of the original Hamiltonian, with the bound of §4. On the lattice,

a volume-uniform gap along a weak-coupling trajectory therefore follows once a scheme carries the flow into the box; the continuum statements T3 and T4 of the obligations map are separate.

Explicit numbers come only from the continuous-time count of the Kogut–Susskind note restricted to adjacent growth (§§2–3). Corrected, that count gives formal values such as $\eta \le 0.104$ for plaquette-like terms ($q = 4$) and $\eta \le 0.00655$ for two-plaquette terms ($q = 8$) at $g^2 = 100$, linear in g^2 and decreasing in q like $2^{-q}/\max(4, q)$. They inherit the missing root estimate, the missing general count and the missing true-vacuum correlation bound of that note's correction, so **no gap follows from them**. Constants explicit; reading labels at each use.

1. The existential box: a hypothesis check of Yarotsky's theorem

Notation as in the T2 note §1: the periodic lattice $\Lambda = (\mathbb{Z}/N_s\mathbb{Z})^3$, each link based at its lower endpoint, $\mathcal{H}_x = L^2(SU(3))^{\otimes 3}$ for the three links based at x, Ω_x the constant function, and units $\hbar c/a = 1$ until restored. Set $\varepsilon = \frac{g^2}{2} \cdot \frac{4}{3} = \frac{2g^2}{3}$, $\varepsilon' = (1 - \eta_E)\varepsilon$, $w_p = \operatorname{tr} U_p + \operatorname{tr} U_p^\dagger$, so $\|w_p\| \le 6$. Write $\Lambda_q = \{-q, \dots, q\}^3$.

Hypotheses of the class $\mathcal{C}_{\mathrm{TI}}(g, \eta, \eta_E, q)$, $q \ge 1$:

- **(E)** $H'_E = \sum_\ell \sum_R e'(R) P_R^\ell$, where P_R^ℓ is the Peter–Weyl isotypic projection of link ℓ, $e'(\mathrm{triv}) = 0$ and $e'(R) \ge \varepsilon'$ for every nontrivial R. The choice $e'(R) = (1 - \eta_E)\frac{g^2}{2} C_2(R)$ is an example.
- **(D)** $D = \sum_j d_j$ with each d_j bounded, self-adjoint, gauge invariant, and acting on a connected set $\operatorname{supp} d_j$ of at most q links; the family is invariant under lattice translations; and $\sum_{j \ni \ell} \|d_j\| \le \eta$ for every link ℓ. The family may depend on N_s.
- **(V)** $N_s \ge 2q + 1$, so every translate $x + \Lambda_q$ embeds in the torus.

Grouping into site terms. Let $n_x(j)$ be the number of links of $\operatorname{supp} d_j$ based at x, and put

$$\delta_x = \sum_j \frac{n_x(j)}{|\operatorname{supp} d_j|}\, d_j.$$

Then $\sum_x \delta_x = D$, because $\sum_x n_x(j) = |\operatorname{supp} d_j|$. The δ_x are translates of δ_0, because the family and the weights are translation covariant. Each j with $n_x(j) > 0$ contains a link based at x, so

$$\|\delta_x\| \le \sum_{i=1}^{3} \sum_{j \ni (x, x+e_i)} \|d_j\| \le 3\eta.$$

A connected set of at most q links containing a link based at x has all its vertices within graph distance q of x, hence in $x + \Lambda_q$; so δ_x acts on $\mathcal{H}_{x+\Lambda_q}$. Under (V), δ_0 lifts to a local term on $\mathbb{Z}^3$ and the interaction is the periodic restriction of a translation-invariant interaction of range Λ_q. Assign each plaquette $p(x, i, j)$, $i < j$, to its corner x, as in the T2 note; its links are based in $x + \{0, 1\}^3 \subseteq x + \Lambda_q$.

Normalized form. Dividing by ε',

$$\frac{H}{\varepsilon'} = \sum_x h'_x + \sum_x \phi_x, \qquad h'_x = \frac{1}{\varepsilon'} \sum_{i=1}^{3} \sum_R e'(R) P_R^{(x, x+e_i)}, \qquad \phi_x = -\frac{1}{\varepsilon' g^2} \sum_{i<j} w_{p(x,i,j)} - \frac{\delta_x}{\varepsilon'}.$$

Every link is based at exactly one site, every plaquette has one assigned corner, and every d_j is shared out with total weight one, so the sums reproduce H/ε'. The hypotheses of Yarotsky's Theorem 1, as recorded in the T2 note §2 (repository hypothesis check read; Yarotsky, Commun. Math. Phys. 261, 799, Section 1 and Theorem 1 at passage level recorded there), are met:

1. h'_x is a function of the product Peter–Weyl partition of unity on $\mathcal{H}_x$, which contains the projection onto Ω_x; $h'_x \Omega_x = 0$ since $e'(\mathrm{triv}) = 0$; on the orthogonal complement of Ω_x some link carries a nontrivial representation, and all terms are nonnegative, so $h'_x \ge \varepsilon'/\varepsilon' = 1$.

2. ϕ_x is a bounded symmetric form on $\mathcal{H}_{x+\Lambda_q}$ with $|\phi_x(v,v)| \le \beta\|v\|^2$ and $\alpha = 0$, where

$$\beta = \frac{3\cdot 6/g^2 + 3\eta}{\varepsilon'} = \frac{1}{1-\eta_E}\left(\frac{27}{g^4} + \frac{9\eta}{2g^2}\right).$$

3. Translation invariance, finite range Λ_q and the periodic cubic volume are as in item 5 of the T2 note.

Proposition (existential target box). Let $\beta_q^* = \beta_*(3,\Lambda_q)$ and $\gamma_q = \gamma(3,\Lambda_q)$ be the constants of Yarotsky's Theorem 1 for $\nu = 3$ and range Λ_q. If $H \in \mathcal{C}_{\mathrm{TI}}(g,\eta,\eta_E,q)$ and

$$\frac{27}{g^4} + \frac{9\eta}{2g^2} \;\le\; (1-\eta_E)\,\beta_q^*, \qquad \text{equivalently} \qquad \eta \;\le\; \frac{2(1-\eta_E)\beta_q^*}{9}\,g^2 - \frac{6}{g^2},$$

then for every $N_s \ge 2q+1$ the operator H on $L^2(SU(3)^{\mathcal{E}})$ has a non-degenerate ground state ψ_0, ψ_0 is gauge invariant, and

$$\Delta_{a,L}^{\mathrm{phys}} \;\ge\; \Delta_{a,L} \;\ge\; \gamma_q\,(1-\eta_E)\,\frac{2g^2}{3}\,\frac{\hbar c}{a},$$

with η in units $\hbar c/a$. The bound is uniform in N_s.

Proof. By items 1–3, Theorem 1 applies to H/ε' with $\alpha = 0$ and $\beta \le \beta_q^*$, volume by volume with the same constants: it gives a non-degenerate ground state and a gap at least γ_q for H/ε', hence at least $\gamma_q\varepsilon'$ for H. Gauge transformations $\mathcal{V}_\gamma$, $\gamma \in SU(3)^\Lambda$, commute with H: P_R^ℓ commutes with left and right translations of U_ℓ, w_p is the trace of a closed holonomy, and each d_j is gauge invariant by (D). Thus $\mathcal{V}_\gamma\psi_0 = c(\gamma)\psi_0$ with c a continuous homomorphism $SU(3)^\Lambda \to U(1)$. Its differential is a Lie-algebra homomorphism into the abelian $i\mathbb{R}$, so it vanishes on commutators; $\mathfrak{su}(3)$ is semisimple and equals its commutator algebra, so the differential is zero, and $c \equiv 1$ on the connected group. The Haar average $P = \int \mathcal{V}_\gamma\,d\gamma$ projects onto the physical subspace and commutes with the resolvent of H, so that subspace reduces H, contains ψ_0, and carries a spectrum contained in that of H. Hence $\Delta_{a,L}^{\mathrm{phys}} \ge \Delta_{a,L}$. Restoring units gives the display. □

The same theorem gives a thermodynamic limit of the ground states with exponentially decaying correlations of local observables (T2 note §2, passage level). The positivity argument of T1 is not used, since a general d_j need not preserve positivity.

Scope. The Proposition covers the qualitative content of the original class $\mathcal{C}(g,\eta,\eta_E,q)$ under three added hypotheses: translation covariance, connected supports (or any support in a fixed translate of a finite set), and a link-local electric term. A general diagonal $E'(\sigma) \ge \varepsilon'|S(\sigma)|$, or perturbations without translation invariance, would need either a version of Yarotsky's theorem for such interactions or the explicit expansion of §2. The dependence of β_q^* and γ_q on q is not recorded in the statement; a decrease with the range is a heuristic expectation, and the 2^{-q} law of §3 is a property of the formal count only.

2. The perturbed continuous-time expansion (formal)

This section keeps the original class $\mathcal{C}(g,\eta,\eta_E,q)$: units $\hbar c/a = 1$, physical sector, $N_s \ge 4$; H_E' gauge invariant, diagonal in a Peter–Weyl basis of the physical subspace, commuting with every support projection P_S, with $E'(\emptyset) = 0$ and $E'(\sigma) \ge \varepsilon'|S(\sigma)|$; each d_j bounded, self-adjoint, gauge invariant, on at most q links, with local norm $\sum_{j\ni\ell}\|d_j\| \le \eta$. Assume each d_j centered, $\langle\Omega_0, d_j\Omega_0\rangle = 0$. Subtracting these constants shifts H by a scalar and at most doubles η. Without it, a term acting away from the excited set contributes an insertion with no energy damping.

Insert the projections P_S of the Kogut–Susskind note §1 between the factors of the Duhamel series for $e^{-t(H_E'-W-D)}$. A history is a sequence of supports $S_0 = \emptyset, S_1, \dots, S_n = \emptyset$ with

insertions X_k, each a plaquette term w_p/g^2 or a term d_j. Since $H'_E \geq \varepsilon'|S|$ on the range of P_S and $\|P_{S'} X P_S\| \leq \|X\|$,

$$|w(\gamma)| \leq \prod_{k=1}^{n} \|X_k\| \, e^{-\varepsilon' A(\gamma)}, \qquad A(\gamma) = \sum_{k=1}^{n-1} |S_k| x_k.$$

Each P_S contains every representation label with support S, so the labels are summed inside the operator norm and need no separate count. Gauge invariance of every insertion keeps the history in the physical sector, so $|S_k| \geq 4$ for $0 < k < n$ (Kogut–Susskind note, §1 correction, $N_s \geq 4$). A d-insertion changes the support only on its own links, so given S_{k-1} there are at most 2^q outcomes S_k. By centering, a d-insertion either touches S_{k-1} or excites a link of its support. The hard-core polymer organization of these histories is assumed as in that note §1; it is proposed there and not established.

Restricted count. Root at a link ℓ_0, and keep the histories whose first insertion contains ℓ_0 and whose later insertions touch the current excited set. Write $h = 6e/g^2$, where the factor e per insertion comes from the term e^n in the Kotecký–Preiss weight $e^{a(\gamma)}$, and $\delta' = \varepsilon' - \lambda = \theta\varepsilon'$ with $\lambda = (1-\theta)\varepsilon'$. A later step has at most $4|S_{k-1}|$ plaquettes touching S_{k-1}, each with 16 outcomes, and the d-terms containing a link of S_{k-1}, with weighted sum at most $|S_{k-1}|\eta$ and 2^q outcomes each. Repeated use of the same plaquette or term is included in these sums. The preceding interval integral is $1/(\delta'|S_{k-1}|)$, so the per-step factor is

$$u' = \frac{64h + e\,\eta\,2^q}{\delta'} = \frac{384e/g^2 + e\,\eta\,2^q}{\delta'}.$$

The first step has the four plaquettes containing ℓ_0, for which S_1 is forced, and the d-terms containing ℓ_0, each with up to 2^q outcomes:

$$F' = 4h + e\,\eta\,2^q = \frac{24e}{g^2} + e\,\eta\,2^q.$$

With n insertions the restricted sum is at most $F' u'^{\,n-1}$, so the majorant with marked-interval multiplier $n-1$ obeys

$$B'_\lambda \;\leq\; F' \sum_{m \geq 1} m\, u'^{\,m} = \frac{F' u'}{(1-u')^2} \;\leq\; 4F'u' \qquad (u' \leq \tfrac{1}{2}).$$

Overlap conditions. In the Kogut–Susskind note §2, each maximal excitation interval begins at an insertion. A d-insertion touches up to q links, so $\sum_\ell r_\ell \leq \max(4,q)\, n$, and the overlap bound becomes $Q'A(\gamma) + \frac{Q'}{4\delta'} \max(4,q)\, n(\gamma)$. Comparing with $a(\gamma) = n + \lambda A$, the sufficient conditions read

$$Q' \leq \lambda, \qquad Q' \leq \frac{4\delta'}{\max(4,q)},$$

where Q' is the all-roots majorant of that section with d-insertions included.

Correction (2026-10-02), status of this section. B'_λ bounds only the restricted class. Q' also contains histories that reach ℓ_0 after their first insertion (the missing roots of the Kogut–Susskind note §3), and histories in which a plaquette or a d-term starts a disjoint piece that is later bridged; the marked general count for these is withdrawn there. Even with Q' controlled, the spectral lemma of that note §4 needs a normalized true-vacuum correlation bound for every bounded gauge-invariant observable, and none is supplied. The former sentence "then the gap is at least 4λ on the cyclic subspace of local observables" is withdrawn: the restricted count proves neither decay on that subspace nor a spectral gap. The rate $4\lambda = \frac{8}{3} g^2 (1-\eta_E)(1-\theta)$ is a candidate rate.

3. The formal numbers

For $SU(3)$ take $\theta = \frac{1}{2}$ and $\eta_E = 0.1$: then $\varepsilon' = 0.6g^2$ and $\lambda = \delta' = 0.3g^2$. Under $u' \le \frac{1}{2}$, $B'_\lambda \le 4F'u' \le 2F'$, and since $\max(4,q) \ge 4$ the binding requirement is $2F' \le 4\delta'/\max(4,q)$. The formal restricted conditions are therefore

$$\text{(R1)} \quad e\,\eta\,2^q \;\le\; 0.15\,g^2 - \frac{384e}{g^2}, \qquad \text{(R2)} \quad e\,\eta\,2^q \;\le\; \frac{0.6\,g^2}{\max(4,q)} - \frac{24e}{g^2}.$$

(R1) is $u' \le \frac{1}{2}$. For $q \le 4$, (R2) reads $e\eta 2^q \le 0.15g^2 - 24e/g^2$ and follows from (R1). For $q = 8$ it reads $e\eta 2^q \le 0.075g^2 - 24e/g^2$. With $384e = 1043.82$, $24e = 65.239$, $16e = 43.4925$ and $256e = 695.880$, the largest η satisfying both, rounded down, in units $\hbar c/a$:

g^2	$q = 4$ (plaquette-like terms)	$q = 8$ (two-plaquette terms)	binding for $q = 8$
100	0.104	0.00655	(R1)
200	0.569	0.0210	(R2)
400	1.31	0.0428	(R2)

For example, at $g^2 = 100$ (R1) gives $(15 - 10.438)/43.4925 = 0.1049$ for $q = 4$ and $4.562/695.880 = 0.006555$ for $q = 8$; at $g^2 = 200$, (R2) for $q = 8$ gives $(15 - 0.326)/695.880 = 0.02109$, below the (R1) value 0.03561. For $\theta \le \frac{1}{2}$ and large g^2 the two conditions together give

$$\eta \;\simeq\; \frac{4\theta(1-\eta_E)\,g^2}{3\max(4,q)\,e\,2^q},$$

which at $\theta = \frac{1}{2}$, $q \le 4$ is the former $g^2(1-\eta_E)/(6e\,2^q)$. **These numbers are formal**: they solve sufficient inequalities for the restricted majorant B'_λ, which does not bound Q', and no gap follows from them (§2).

The general factor $24u$ of the former Kogut–Susskind count is withdrawn in that note §3, and with it the remark that the table holds at 24 times the coupling. As algebra, $24u' \le \frac{1}{2}$ reads $e\eta 2^q \le 0.00625g^2 - 384e/g^2$, which has no solution $\eta \ge 0$ unless $g^4 \ge 61440e \simeq 1.670 \times 10^5$, that is $g^2 \ge 408.7$. Tolerances "divided by 24 at the listed couplings" therefore never existed.

4. What the box asks of the intermediate region

A renormalization step at scale a produces an effective Hamiltonian at $2a$. Suppose a scheme exists such that

1. each step is an exact low-energy reduction: $E_0(a) < E_c(a)$, and below $E_c(a)$ the spectrum of $H(a)$, with multiplicities and after a common constant shift, coincides with that of $H_{\text{eff}}(2a)$;
2. the effective Hamiltonian stays in the class $\mathcal{C}_{\text{TI}}(g(a), \eta(a), \eta_E(a), q)$ with q fixed;
3. the coupling runs, $g(2a) > g(a)$, and after finitely many steps, at a scale a_s, reaches the region of the Proposition: $27/g_s^4 + 9\eta_s/(2g_s^2) \le (1 - \eta_{E,s})\beta_q^*$.

Here Δ is the gap in the sector on which the reduction acts (full or physical; the Proposition bounds both). Item 1 gives $\Delta(H(a)) \ge \min\{\Delta(H_{\text{eff}}(2a)),\, E_c(a) - E_0(a)\}$: if $E_1(a) < E_c(a)$ the two lowest levels are shared and the gaps agree; otherwise the gap is at least $E_c(a) - E_0(a)$. Iterating from $a_0 = a_{\text{UV}}$ through $a_k = 2^k a_0$,

$$\Delta(H(a_{\text{UV}})) \;\ge\; \min\Big\{\gamma_q(1-\eta_{E,s})\frac{2g_s^2}{3}\frac{\hbar c}{a_s},\; \min_{k<s}\big(E_c(a_k) - E_0(a_k)\big)\Big\},$$

uniformly in the volume over lattices whose final side is at least $2q+1$. This is positive in physical units when, as $a_{\mathrm{UV}} \to 0$, a_s stays bounded above and the margins $E_c - E_0$ stay bounded below in physical units; that requirement is part of T3. Under items 1–3 at every scale of a trajectory, the bare couplings on it have positive volume-uniform gaps, the content of T2′ there. Items 1–3 in the weak region have analogues in the constructive programme (Balaban's small-field effective actions are the Euclidean analogue of item 2); in the intermediate region they are the open problem.

The box makes the finish line definite and existential: the flow has to arrive at couplings and correction sizes satisfying item 3. The formal count of §3 suggests, without proof, $g^2 \sim 10^2$ with corrections of local norm about $10^{-1}\,\hbar c/a$ for single-plaquette terms and below $10^{-2}\,\hbar c/a$ for two-plaquette terms. The repository's heuristic extraction, $\beta \simeq e^{-465}$ for the range $\{0,1\}^3$, is a heuristic proof extraction (Kogut–Susskind note, §5); β_q^* for the larger range Λ_q has not been extracted.

The one-loop distance from $g^2 = 1/2$ to $g^2 = 100$ is $(2 - 0.01)/0.0966 \simeq 21$ doublings, using $1/g^2(2a) \simeq 1/g^2(a) - 2b_0 \log 2$ with $2b_0 \log 2 = 0.0966$ for $SU(3)$ (heuristic count). The running is perturbative only at the start of that stretch, $g^2 \sim 10^2$ is the formal target of §3, and how the corrections behave over the rest of it is the open question.

5. Consequence for STATE

The note's proved output is the existential target box of §1, a hypothesis check of Yarotsky's theorem: translation-invariant, connected, gauge-invariant terms on at most q links with local norm $\eta \le \frac{2(1-\eta_E)\beta_q^*}{9} g^2 - \frac{6}{g^2}$ (units $\hbar c/a$), with a link-local electric term, leave a non-degenerate gauge-invariant ground state and a volume-uniform full and physical gap of at least $\gamma_q(1 - \eta_E)\frac{2}{3}g^2\,\hbar c/a$, with β_q^*, γ_q existential. The location of the box is not explicit. The explicit table of §3 and the target $g^2 \sim 10^2$ are formal restricted estimates that inherit the gaps named in the Kogut–Susskind correction; downstream uses of the box location, including abelian comparison and intermediate-region-finite-verification, should read it existentially. An explicit box needs a complete marked general-history count with q-link insertions and a normalized true-vacuum correlation bound, or explicit Yarotsky constants for range Λ_q. This correction changes no other note and no STATE file.